Example 6.24

The probability density function of a random variable X is

$$f(x)=ke^{-|x|},-\infty<x<\infty$$

Find the value of k and also find mean and variance for the random variable.

Solution

We know that,

$$\int_{-\infty}^{\infty} f(x)dx=1$$

$$\int_{-\infty}^{\infty} k\,e^{-|x|}dx=1$$

$$k\int_{-\infty}^{\infty} e^{-|x|}dx=1$$

$$2k\int_{0}^{\infty} e^{-x}dx=1$$

($\because$ $e^{-|x|}$ is an even function)

$$2k\left[\frac{e^{-x}}{-1}\right]_0^{\infty}=1$$

$$k=\frac{1}{2}$$

Mean of the random variable is

$$E(X)=\int_{-\infty}^{\infty} x\,f(x)dx$$

$$E(X)=\int_{-\infty}^{\infty} x\,k\,e^{-|x|}dx$$

($\because$ $xe^{-|x|}$ is an odd function of x)

$$=\frac{1}{2}\int_{-\infty}^{\infty} x\,e^{-|x|}dx$$

$$=0$$

$$E\left(X^2\right)=\int_{-\infty}^{\infty} x^2 f(x)dx$$

$$=\int_{-\infty}^{\infty} x^2 k\,e^{-|x|}dx$$

$$=\frac{1}{2}\int_{-\infty}^{\infty} x^2 e^{-|x|}dx$$

$$=\int_{0}^{\infty} x^2 e^{-x}dx \quad (\because \ e^{-|x|} \text{ is an even function})$$

$$=\Gamma(3)\left(\because \Gamma(\alpha)=\int_0^{\infty} x^{\alpha-1}e^{-x}dx,\alpha>0;\Gamma(n)=(n-1)!\right)$$

$$=2$$

$$V(X)=E\left(X^2\right)-\left[E(X)\right]^2$$

$$=2-[0]^2$$

$$=2$$

Exercise 6.2

1. Find the expected value for the random variable of an unbiased die
2. Let X be a random variable defining number of students getting A grade. Find the expected value of X from the given table

$X=x$	0	1	2	3
$P(X=x)$	0.2	0.1	0.4	0.3

3. The following table is describing about the probability mass function of the random variable X

x	3	4	5
$P(x)$	0.2	0.3	0.5

Find the standard deviation of x.

4. Let X be a continuous random variable with probability density function

$$f_X(x)=\begin{cases}2x, & 0\le x\le 1\\ 0, & otherwise\end{cases}$$

Find the expected value of X.

5. Let X be a continuous random variable with probability density function

$$f(x)=\begin{cases}\frac{3}{x^4}, x\ge 1\\ 0, \quad otherwise\end{cases}$$

Find the mean and variance of X.

6. In an investment, a man can make a profit of ₹ 5,000 with a probability of 0.62 or

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